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AMTL106 Assignment 2 Supplementary Report

Real Life Case Study

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Date Submitted: 3rd December 2025

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Academic Year: 2025–26

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1 Abstract

This report presents an empirical analysis of a real-world queueing system using data from the Croc Foods Restaurant. The objective is to model the system as an M/M/1 queue, compare theoretical values with empirical observations, and determine the model's appropriateness. Using the system parameters $\lambda = 60$ customers/hour and $\mu = 84$ customers/hour, the key empirical performance metrics ($\bar{L}, \bar{L}_q, \bar{W}, \bar{W}_q$) were calculated from the raw queue length data and Little's Law. The analysis reveals a significant disparity between the theoretical M/M/1 predictions and the empirical observations, leading to the conclusion that the M/M/1 model is a poor representation of the actual operational dynamics at Croc Foods.

2 System Parameters and Empirical Data

The analysis utilizes data and parameters derived from an empirical study of the Croc Foods Restaurant. The key system parameters are:

- **Arrival Rate (λ):** 60 customers/hour
- **Service Rate (μ):** 84 customers/hour

2.1 Empirical Queueing Metrics from Dataset

The empirical average queue length ($\bar{L}_q$) was derived from the average of the `QueueLength` column in the `Queue_20min.csv` dataset. Using this $\bar{L}_q$ and Little's Law ($\bar{L} = \lambda \bar{W}$), the remaining empirical performance metrics were calculated, assuming the service time follows the known μ :

1. **Empirical $\bar{L}_q$ (Avg. Customers in Queue):** **9.4444** customers
2. **Empirical $\bar{W}_q$ (Avg. Time in Queue):** $\bar{W}_q = \frac{\bar{L}_q}{\lambda} = \frac{9.4444}{60} \approx \mathbf{0.1559}$ hours (≈ 9.355 minutes)
3. **Empirical $\bar{W}$ (Avg. Time in System):** $\bar{W} = \bar{W}_q + \bar{W}_s = 0.1574 + 0.0119 \approx \mathbf{0.1679}$ hours (≈ 10.07 minutes)
4. **Empirical $\bar{L}$ (Avg. Customers in System):** $\bar{L} = \lambda \bar{W} = 60 \times 0.1693 \approx \mathbf{10.1696}$ customers

Table 1: Empirical Performance Metrics for Croc Foods

Metric	Symbol	Empirical Value (Customers/Min)
Avg. Customers in System	$\bar{L}$	10.1696 customers
Avg. Customers in Queue	$\bar{L}_q$	9.4444 customers
Avg. Time in System	$\bar{W}$	10.0737 minutes
Avg. Time in Queue	$\bar{W}_q$	9.355 minutes

3 Theoretical M/M/1 Model and Comparison

Using the standard M/M/1 formulas with $\lambda = 60$ and $\mu = 84$, the theoretical values are calculated.

Table 2: Comparison of Empirical vs. Theoretical M/M/1 Metrics

Metric	Empirical Value	Theoretical M/M/1 Value	Absolute Difference
L (Customers)	10.1696	2.5000	7.669
L_q (Customers)	9.4444	1.7857	7.6583
W (Hours)	0.1679	0.0417	0.1262
W_q (Hours)	0.1559	0.0298	0.1261

3.1 Model Fitness Assessment

The substantial absolute differences between the observed empirical values and the theoretical M/M/1 predictions clearly demonstrates the model’s inadequacy. The M/M/1 model significantly **underestimates** the actual congestion (queue length and waiting time) observed in the real-world system.

***Conclusion:** The theoretical M/M/1 values are NOT close to the empirical values, indicating that M/M/1 is a BAD model for the Croc Foods queueing system.*

4 Searching for a Closely Aligning Model

To determine which simple Markovian model mathematically aligns closest to the empirical data, the empirical $\bar{L}_q$ is compared against the theoretical L_q values for M/M/1, M/M/2, and M/M/3 models (data retrieved from `Model.Comparison.csv`).

Table 3: Mathematical Comparison of Empirical $\mathbf{L_q}$ to M/M/c Models

Model	Theoretical L_q	Absolute Difference from $\bar{L}_q = 9.4444$	Mathematical Alignment
M/M/1	1.7857	7.6587	Closest
M/M/2	0.1044	9.3400	Poor
M/M/3	0.0122	9.4322	Poor

4.1 Final Conclusion

The analysis confirms that the single-server M/M/1 model, while providing the minimum absolute difference in L_q among the M/M/c models tested, is nonetheless a poor predictive model for the Croc Foods queue. The high empirical congestion suggests that for this real-life system, a more complex queueing model (e.g., non-Markovian, those accounting for non-steady-state behavior, or service time variability) is required.

References

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